# Current Affairs 13-11-2026

## Credit - Drishti Ias

# NITI Aayog Report on Impact Assessment of PLI Schemes

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**For Prelims: NITI Aayog , Production Linked Incentive (PLI) scheme , Electronics Component Manufacturing Scheme , PLI Scheme for Medical Devices , Telecom PLI Scheme , White Goods , High-Efficiency Solar PV Modules**

**For Mains:** Significance of Production Linked Incentive (PLI) Scheme for manufacturing sector, Atmanirbhar Bharat, Make in India.

**Source: PIB**

## Why in News?

**NITI Aayog** , with **IIM Ahmedabad** as its knowledge partner, has published a mid-course report titled **“Impact Assessment of PLI Schemes”** . It evaluates all **14 Production Linked Incentive (PLI) schemes** on four outcomes: **investment, production, employment and exports** .


## Summary

- **PLI schemes have strengthened India's manufacturing ecosystem** , mobilising **₹2.40 lakh crore in investments** , generating **₹22.66 lakh crore in production** and creating **8.49 lakh jobs** across 14 sectors as of March 2026.
- **Key challenges include low employment generation, limited domestic value addition and uneven sectoral performance.** Strengthening demand, local supply chains and Centre-State coordination is essential for achieving **Viksit Bharat @2047** .


## What are the Key Highlights of the “Impact Assessment of PLI Schemes”?

### Macro-Economic Impact

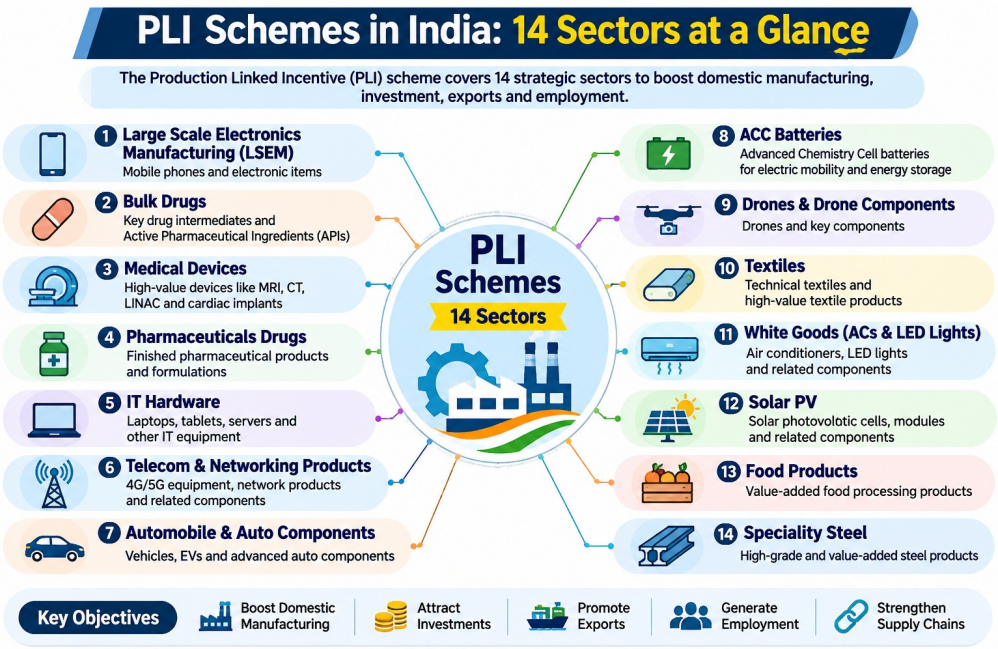
- **Overall Macro-Economic Performance (till March 2026)** : The report utilizes a 50% threshold of **full-tenure targets as a mid-course benchmark** to evaluate the schemes.
- **Investment Mobilisation:** Cumulative investment reached **Rs 2.40 lakh crore** , achieving **76% of the full-tenure target** of **Rs** 3.15 lakh crore.
- **Production Growth:** Cumulative production/sales stood at **Rs 22.66 lakh crore** , achieving **64% of the full-tenure target** of **Rs** 35.37 lakh crore.
- **Employment Generation:** Cumulative employment reached **8.49 lakh jobs** , representing **28% of the full-tenure target** of 30.19 lakh jobs.
- **Implementation Progress:** The 14 PLI schemes have completed **58% of their aggregate implementation period** , with **42% of the implementation period still remaining** .
- **Incentive Disbursement:** Cumulative disbursement stood at **Rs 35,354 crore as of March 2026** , accounting for **18.51% of the total financial outlay of Rs 1.91 lakh crore** .
  - Incentives are released based on the fulfilment of prescribed investment and production conditions, rather than upfront.
- **Gestation Period:** Employment generation may take longer because manufacturing investments are followed by capacity commissioning, production ramp-up and expansion of commercial operations.

## Sector-Wise Performance of PLI Scheme

- **Sectoral Progress:** Nearly 70% of targeted PLI segments are broadly on track, with particularly strong outcomes in electronics, telecom and pharmaceuticals, although performance varies across sectors.
- **Large Scale Electronics Manufacturing (LSEM):** The **LSEM PLI** scheme has emerged as one of the most efficient schemes in terms of resource utilisation and output generation, with mobile exports rising nearly nine-fold to Rs 2 lakh crore in FY2025 and **imports declining by 77%.**
  - It has strengthened domestic manufacturing and paved the way for the **Electronics Component Manufacturing Scheme** and the **Mobile Phone Manufacturing Scheme** to deepen domestic value addition.
- **Pharmaceutical:**
  - **Bulk Drugs:** The **PLI Scheme for Bulk Drugs** has strengthened **domestic API manufacturing and reduced import dependence** , helping India transition from a **Rs 1,930 crore trade deficit (FY2021–22) to a Rs 2,280 crore surplus (FY2024–25)** .
    - It has created **55,000 MT manufacturing capacity across 26 critical APIs** .
    - Chemical synthesis-based products remain the strongest performers. The scheme has also boosted investment, exports, employment and pharmaceutical self-reliance.
  - **Manufacturing of Medical Devices:** The **PLI Scheme for Medical Devices** has boosted domestic manufacturing of high-value medical equipment, including **MRI machines , CT scanners, LINACs , cardiac implants and dialysers, reducing import dependence.**
  - **Pharmaceutical PLI Scheme:** The **PLI Scheme for Pharmaceuticals (Rs 15,000 crore outlay)** has attracted **over Rs 45,000 crore in investments** through 55 approved applicants and enabled domestic manufacturing of **191 pharmaceutical products for the first time** .
    - Pharmaceutical exports increased **32.7%** , from **USD 19.46 billion (2021) to USD 25.82 billion (2025)** .
    - Large firms contributed **61% of total production** , while MSMEs and mid-sized firms recorded moderate growth.
- **PLI Scheme for IT Hardware 2.0:** It aims to expand India's electronics manufacturing beyond mobile phones.

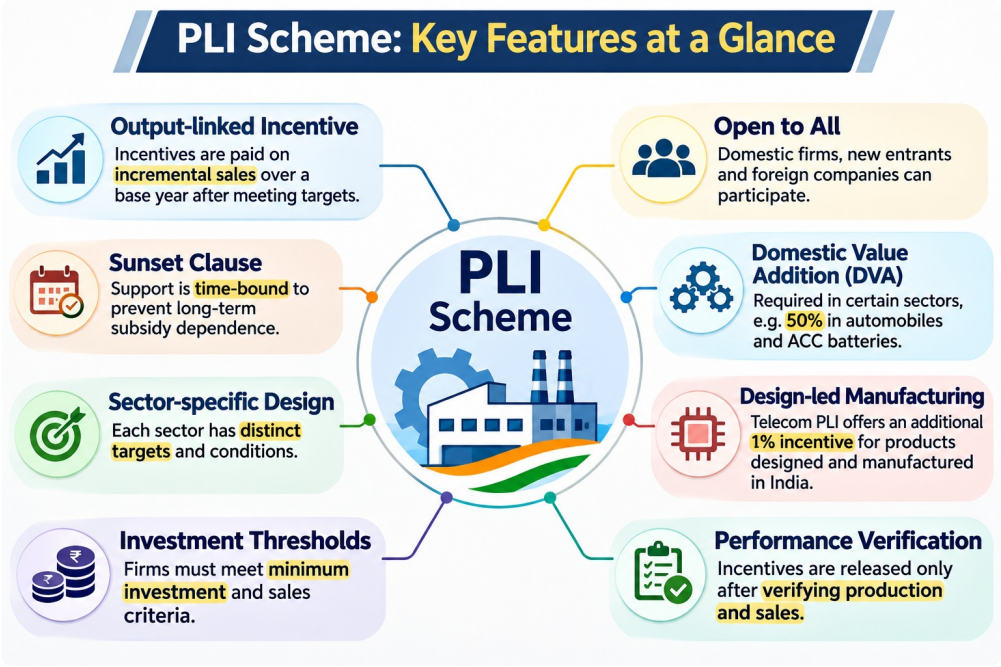
- The earlier **IT Hardware 1.0 underperformed due to low incentives (1–4%, averaging 2%) a** nd stringent localisation requirements. The revised scheme offers approximately 5% average incentives.

- **Telecom & Networking Products:** The **Telecom PLI Scheme** has strengthened domestic 4G/5G equipment manufacturing, with telecom exports growing at **1.5% AAGR** and imports declining by **18.5% average annual growth rate (AAGR) (FY21–FY25)** .
  - It provides an additional **1% incentive for design-led manufacturing** .
- **Automobile and Auto Components:** The **PLI Scheme for Automobiles and Auto Components** is accelerating India's transition towards electric mobility and advanced automotive technologies.
  - Vehicle exports increased **33.2%, from USD 18.9 billion (2021) to USD 25.16 billion (2025).**
  - The scheme mandates 50% Domestic Value Addition (DVA) and follows an open-to-all approach.
- **Advanced Chemistry Cell (ACC) Battery:** The **PLI Scheme for ACC Batteries** aims to build domestic battery manufacturing capacity for electric mobility and energy storage.
  - However, progress is severely delayed due to stringent localisation norms, long gestation period, technology gaps and dependence on critical minerals.
- **Drones & Drone Components:** The **Drone PLI Scheme** has boosted domestic drone manufacturing, with sector turnover increasing seven-fold. It has disbursed 77.5% of its outlay, strengthening India's indigenous drone manufacturing and component ecosystem.
- **Textile Products:** The **PLI Scheme for Technical Textiles** has boosted high-value manufacturing, with **imports declining by 49.4%** and **exports increasing by 59% (2021–2025)** .
  - However, **fund utilisation remains low at 3.61%** , partly due to high eligibility thresholds and rigid **HSN codes.**
- **White Goods:** The **PLI Scheme for White Goods** has boosted domestic manufacturing and localisation of AC and LED components.
  - Compressor manufacturing capacity **increased ten-fold, from 1 million units (2021) to 10 million units (2025–26),** while investments across 18 states strengthened the domestic supplier ecosystem. **However, low incentive disbursement remains a concern.**
- **High Efficiency Solar PV Modules:** The **PLI Scheme for High-Efficiency Solar PV Modules** has accelerated India's solar manufacturing, with **21% of domestic capacity creation directly attributable to the scheme** and declining module imports.
  - It has promoted integrated manufacturing across polysilicon, wafers, cells and modules.
  - However, technology gaps, volatile global prices and dependence on concentrated international supply chains remain key challenges.
- **Food Products:** The **PLI Scheme for Food Processing** has boosted sectoral growth, with sales rising to **Rs 1.09 lakh crore in FY2025–26** , generating **3.29 lakh jobs** and creating **35 lakh MT annual processing capacity** .
  - It has strengthened agricultural value chains and supply chains. However, **weak global branding support and limited evidence of creating globally competitive firms** remain concerns.
- **Speciality Steel:** The **PLI Scheme for Specialty Steel** aims to boost domestic production of high-value steel for automobiles, defence, energy and infrastructure.
  - It covers **67 applicants across 21 sub-categories** , with production showing an upward trend. However, **frequent policy revisions, implementation delays and production below expectations** remain key concerns.




# What is the Production Linked Incentive (PLI) Scheme?

- **About:** Launched in 2020 as a central pillar of **Atmanirbhar Bharat** , the PLI scheme is a performance-based industrial policy spanning 14 sectors with an outlay of Rs 1.97 lakh crore, offering financial incentives tied strictly to **incremental sales** over a base year to domestic and foreign manufacturers.
  - It was introduced after **Covid-19** exposed India’s import dependence and disrupted global supply chains.
  - It also aims to use the **China+1 shift** in global supply chains to attract investment.
  - As of November 2026, PLI covers **14 sectors** with a total outlay of about **Rs 1.91 lakh crore** .
- **Objectives:**
  - **Scale and Competitiveness:** To achieve **economies of scale** and enhance the global competitiveness of Indian manufacturing firms.
  - **Integration into GVCs:** To seamlessly integrate India’s domestic industries into complex **Global Value Chains (GVCs)** .
  - **Strategic Import Substitution:** To drastically reduce import dependence in critical sectors like **APIs** , semiconductors, and electronic components.
  - **Employment Generation:** To spur massive job creation, particularly in **labour-intensive sectors** , accelerating the vision of **Viksit Bharat @2047.**
- **Three Strategic Segments of PLI:**
  - **Labour-Intensive Sectors:** Focuses on massive job creation and leveraging India's demographic dividend (e.g., **Textiles, Food Processing** ).
  - **Value-Chain Integrated Sectors:** Target industries requiring deep supply chain networks and high-volume output (e.g., **Large-Scale Electronics, Automobiles, Pharmaceuticals, White Goods** ).
  - **Sunrise Sectors:** Aims at securing early-mover advantages in next-generation, critical technologies (e.g., **Advanced Chemistry Cell (ACC) Batteries, Solar PV Modules, Drones** ).
- **Key Features of the PLI Scheme:**




## What are the Measures Recommended by the NITI Aayog to Strengthen PLI Schemes?

- **Adopt a Sector-Specific Approach:** Move beyond a uniform approach by studying the unique constraints of different sectors (such as technology adoption, gestation periods, and capital requirements) to make future policy interventions more strategically targeted.
- **Integrate Demand-Pull Mechanisms:** Balance the current supply-driven (production-focused) model with demand-side support.
  - This includes building stronger domestic procurement frameworks, stimulating domestic consumption of high-value goods, and developing export markets.
- **Standardise Data Collection:** Establish uniform reporting mechanisms for **investment, production, employment, exports and additional revenue** to improve monitoring and evaluation.
- **Enhance Industry and Centre-State Consultation:** Involve industry in setting realistic targets and strengthen coordination between **Central and State Governments** to address regulatory and implementation barriers.
- **Ensure Long-Term Policy Support:** Account for long gestation periods and avoid premature assessment of outcomes, particularly in technology-intensive sectors.
- **Promote Global Competitiveness:** Focus on **technology transfer, domestic value addition, export competitiveness and integration into global value chains** to achieve the vision of **Viksit Bharat 2047** .

## Conclusion

The **PLI Scheme has shifted India towards performance-linked, sector-specific industrial support** , with investment and production exceeding implementation progress. However, **employment generation, exports and domestic value addition remain areas of concern** . Its long-term success depends on **demand creation, stronger local supply chains and State-level coordination** to achieve **Viksit Bharat @2047** .

***Drishti Mains Question:***

Examine how the PLI scheme can help India capitalise on the China+1 strategy and emerge as a global manufacturing hub.

## Frequently Asked Questions (FAQs)

**1. What is the Production Linked Incentive (PLI) Scheme?**
Launched in **2020** under Atmanirbhar Bharat, PLI provides financial incentives to eligible manufacturers based on incremental sales over a specified base year.

**2. How many sectors are covered under the PLI scheme?**
The scheme covers **14 strategic sectors** , with a total financial outlay of approximately **Rs 1.91 lakh crore** , according to the NITI Aayog assessment.

**3. What are the major objectives of PLI schemes?**
To promote **domestic manufacturing, economies of scale, employment generation, import substitution, exports and integration into Global Value Chains** .

**4. What are the key findings of the NITI Aayog assessment?**
As of March 2026, PLI schemes mobilised **Rs 2.40 lakh crore in investment** , generated **Rs 22.66 lakh crore in production** and created **8.49 lakh jobs** .

**5. What measures has NITI Aayog recommended to improve PLI schemes?**
It recommends **sector-specific policy design, demand-side support, standardised data collection, Centre-State coordination and long-term policy support** .

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/wocQBda9ML8** ]

## UPSC Civil Services Examination, Previous Year Questions (PYQs)

### *Prelims :*

**Q. Consider, the following statements : (2023)**

**Statement-I :** India accounts for 3.2% of global export of goods.

**Statement-II :** Many local companies and some foreign companies operating in India have taken advantage of India's 'Production-linked Incentive' scheme.

**Which one of the following is correct in respect of the above statements?**

**(a)** Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I

**(b)** Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I

**(c)** Statement-I is correct but Statement-II is incorrect

**(d)** Statement-I is incorrect but Statement-II is correct

**Ans: (d)**

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### *Mains:*

**Q.** Economic ties between India and Japan while growing in the recent years are still far below their potential. Elucidate the policy constraints which are inhibiting this growth. **(2013)**

**Q.** Account for the failure of manufacturing sector in achieving the goal of labour-intensive exports. Suggest measures for more labour-intensive rather than capital-intensive exports **(2017)**

**Q.** The nature of economic growth in India in recent times is often described as a jobless growth. Do you agree with this view? Give arguments in favour of your answer. **(2015)**

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# Agrivoltaics: Integrating Agriculture with Solar Energy in India

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**Source: PIB**

## Why in News?

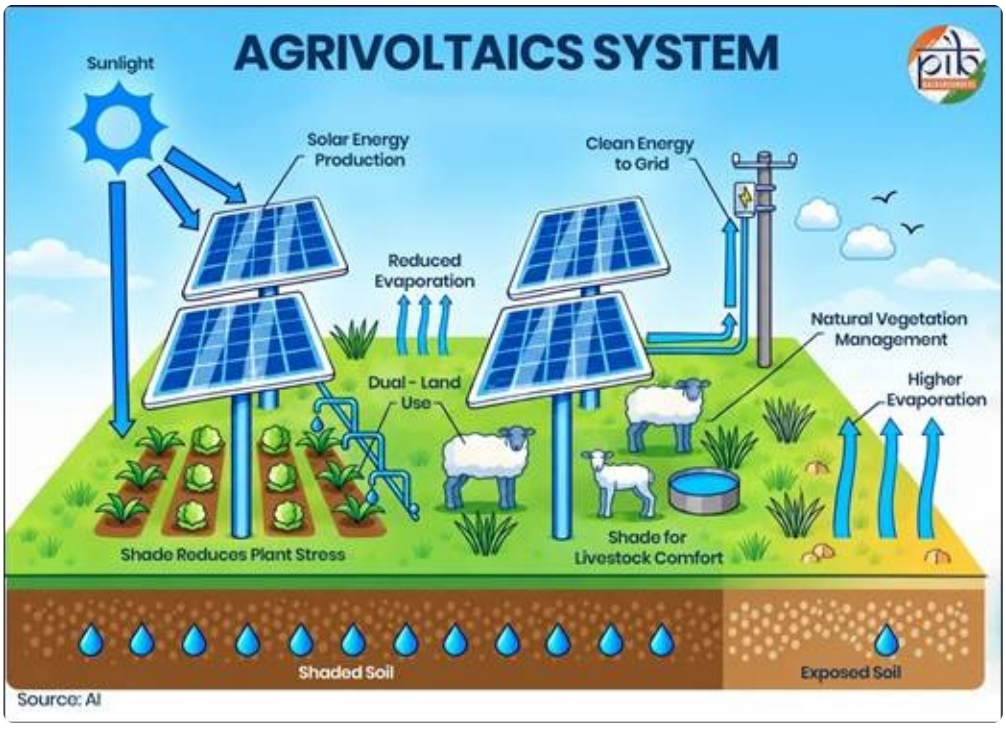
The **Bharat Renewable Energy Summit & Expo 2026 , held at Bharat Mandapam, New**

**Delhi,** highlighted India's efforts to accelerate renewable energy adoption, investment and technological innovation.

- In this context, **agrivoltaics (AgriPV)** has gained relevance as a promising approach to integrate **solar power generation with agricultural production, improve land-use efficiency and diversify farmers' incomes.**

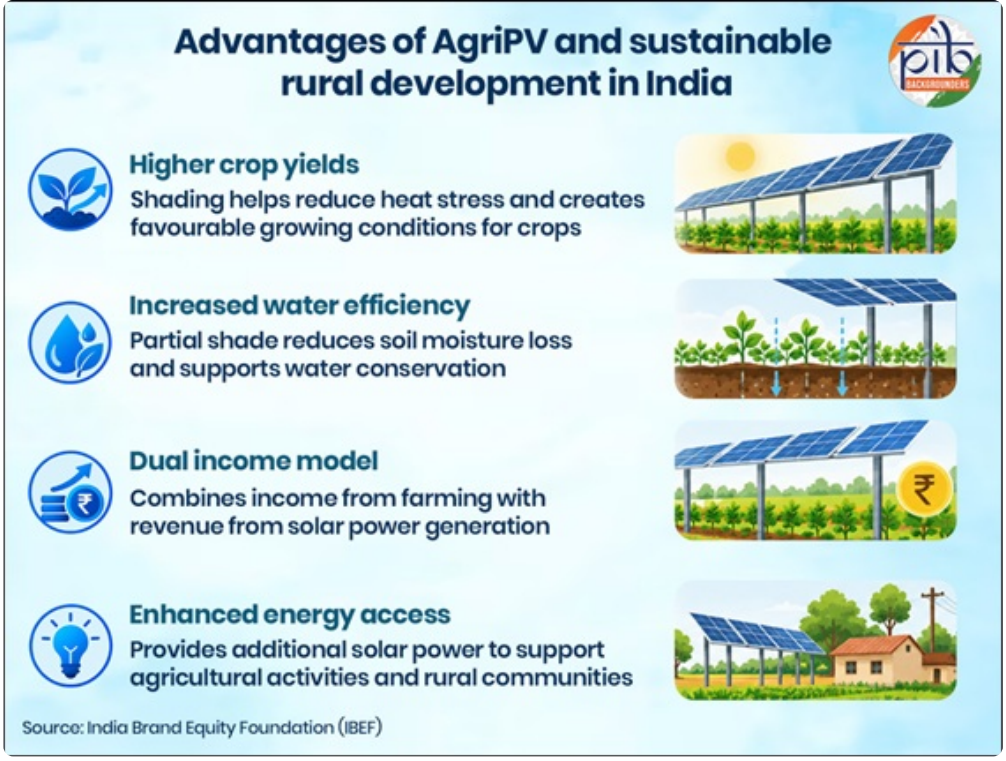
# What are the Key Facts About Agrivoltaics?

- **About:** Agrivoltaics, also known as **agrisolar or dual-use solar, refers to the simultaneous use of agricultural land for crop cultivation** or livestock rearing and solar photovoltaic electricity generation.
- **Working Principle:** Solar panels are **installed above crops or between crop rows,** allowing electricity generation while retaining agricultural activities on the same land.
- **Major Components:**
  - **Elevated Solar Structures:** Solar Panels are mounted at a **suitable** height and spacing to permit crop growth, sunlight penetration, rainfall access and movement of agricultural workers and machinery.
  - **Partial Shading:** Solar panels provide controlled shade, which can **reduce crop heat stress and soil moisture evaporation** under suitable climatic and agronomic conditions.
  - **Dual Land Productivity:** The same land produces agricultural output and solar electricity, improving overall land-use efficiency and potentially generating additional farm income.
- **Applications:** Electricity generated can **power irrigation pumps, cold storage, farm machinery and other agricultural operations** , while surplus electricity may be sold to the grid under applicable arrangements.
- **Crop Compatibility:** Fruits, vegetables, spices and **medicinal plants offer promising opportunities,** depending on their sunlight requirements, local climate and solar installation design.
- **Distinction from Conventional Solar Farming:** Conventional ground-mounted solar installations generally prioritise electricity generation, whereas **agrivoltaics is designed to retain meaningful agricultural production** alongside solar power generation.



# What is the Status and Potential of Agrivoltaics in India?

- **Agriculture and Energy Needs:** Agriculture and allied activities employ approximately **46.1% of India's workforce** , while the sector's Gross Value Added (GVA) is estimated at **₹52.08 lakh crore in 2025–26** . Reliable and affordable energy is essential for irrigation and other agricultural operations.
- **Solar Potential in Agriculture:** According to **Land Use Statistics 2023–24** , India's reported geographical area is **306.96 million hectares** , of which approximately **55% is agricultural land** . The net sown area is **138.99 million hectares** , while the gross cropped area is **217.88 million hectares** , offering considerable scope for integrating solar energy with farming.
- **Current Deployment:** As of August 2025, India had **36 operational agrivoltaic projects with a combined installed capacity of 37.54 MW** , alongside **10 pilot projects under development** .
- **Diverse Models:** Projects across different agroclimatic regions employ **elevated solar structures, interspace layouts** and solar tracking systems, reflecting evolving approaches to integrating agricultural production with solar electricity generation.
- **Large Technical Potential:** Estimates place India's agrivoltaic potential between **3,000 GW and nearly 14,000 GW** , indicating considerable scope for combining renewable energy generation with agriculture.
- **Additional Farm Income:** Agrivoltaics can diversify farmers' income through agricultural production and electricity generation. Under favourable conditions, annual earnings could increase from approximately **₹60,000 per acre to over ₹1 lakh per acre** , although actual returns depend on crop productivity, installation costs and electricity tariffs.
- **Land Suitability:** GIS -based national screening identified **47.35 million hectares of restricted cropland** as potentially suitable for AgriPV after applying environmental, agronomic and technical filters.
- **High-Potential States: Maharashtra, Rajasthan, Andhra Pradesh, Madhya Pradesh, Karnataka and Uttar Pradesh** collectively account for nearly **50% of India's estimated AgriPV power generation potential** , supported by favourable cropland availability, solar resources and agroclimatic conditions.
- **Moderate-Potential States:** Odisha, Tamil Nadu, **Kerala, Telangana, Bihar and West Bengal** also offer opportunities for agrivoltaics development.
- **Regional Advantage:** Southern, western and central India offer favourable combinations of **solar irradiance, cropland availability and horticultural activity,** creating opportunities for large-scale agrivoltaics deployment.




# What are the Major Government Initiatives Promoting Agrivoltaics?

## Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan (PM-KUSUM)

- **About:** Launched **in February 2019,** **PM-KUSUM** promotes solar power adoption in agriculture, reduces dependence on diesel-powered irrigation and creates opportunities for supplementary farm income.
- **Financial Assistance:** The scheme envisages **Central financial assistance of approximately ₹34,422 crore.**
- **Major Components:**
  - **Component A:** Installation of decentralised, grid-connected solar power plants on eligible agricultural or other permitted land.
  - **Component B:** Installation of standalone solar-powered agricultural pumps.
  - **Component C:** Solarisation of existing grid-connected agricultural pumps, including individual and feeder-level solarisation.
- **Relevance for Agrivoltaics:** Supports the broader ecosystem of decentralised solar generation, agricultural electricity supply and farmer participation in clean energy.

## Centre of Excellence on Agrivoltaics

- **Institutional Collaboration:** Established by the **National Institute of Solar Energy (NISE)** in collaboration with the **Indian Council of Agricultural Research–Indian Agricultural Research Institute (ICAR-IARI).**
- **Objectives:** Undertakes **agrivoltaic research, technology demonstration, capacity building and assessment** of crop–solar compatibility.
- **Demonstration Project:** Includes a **100-kW agrivoltaic installation** at the Indian Agricultural Research Institute.

## India Agrivoltaics Alliance (IAA)

- **About:** Launched in 2023, the **IAA is a multi-stakeholder platform anchored by the National Solar Energy Federation of India (NSEFI),** comprising over 40 members.
- **Promotes policy advocacy, research collaboration, knowledge exchange** and pilot projects to accelerate agrivoltaics adoption.
- Supports the **development of scalable, farmer-centric business models integrating agricultural production** with solar electricity generation.

## Major State-Level Initiatives Supporting Solar Energy in Agriculture

- **Maharashtra (Mukhyamantri Saur Krushi Vahini Yojana 2.0):** Seeks to expand agricultural solar electricity supply through approximately 16,000 MW of solar projects on government and private land.
  - Supports daytime power availability for agricultural consumers, potentially facilitating wider integration of solar energy with farming.
- **Gujarat (Kisan Suryoday Yojana):** Promotes daytime electricity supply for agriculture, supporting irrigation and reducing dependence on night-time power availability.
  - Provides for phased expansion of agricultural electricity availability, with a planned supply window extending from 5 AM to 9 PM.
- **Karnataka (Surya Raitha Scheme):** Encourages the use of solar-powered irrigation pumps, allowing eligible farmers to utilise electricity for **irrigation and potentially earn income from surplus power supplied to the grid.**
  - Supports renewable energy adoption and reduced diesel consumption in agriculture.

# Conclusion

Agrivoltaics offers India a promising pathway to integrate **agricultural productivity with renewable energy generation** , improving land-use efficiency, diversifying farmers' incomes and supporting climate-resilient agriculture. However, its large-scale adoption requires **crop-compatible designs, affordable financing, scientific validation, clear regulations and active farmer participation** . Strengthening these foundations can help India achieve a sustainable balance between **food security, energy security and rural development** .

***Drishti Mains Question***

Agrivoltaics presents an opportunity to address the competing demands of agricultural land and renewable energy development. Discuss.

# Frequently Asked Questions (FAQs)

**Q1. What is agrivoltaics?**
Agrivoltaics is the simultaneous use of agricultural land for crop cultivation or livestock rearing and solar photovoltaic electricity generation, improving land-use efficiency.

**Q2. What is India's estimated agrivoltaic potential?**
India's estimated technical agrivoltaic potential ranges from approximately 3,000 GW to 14,000 GW, depending on land suitability and technological assumptions.

**Q3. How many operational agrivoltaic projects are there in India?**
As of August 2025, India had 36 operational agrivoltaic projects with a combined capacity of 37.54 MW, alongside 10 pilot projects under development.

**Q4. Which States have the highest agrivoltaic potential?**
Maharashtra, Rajasthan, Andhra Pradesh, Madhya Pradesh, Karnataka and Uttar Pradesh collectively account for nearly 50% of India's estimated agrivoltaic generation potential.

**Q5. What is the role of PM-KUSUM in promoting agrivoltaics?**
PM-KUSUM supports decentralised solar power generation and solarisation of agricultural pumps, creating an enabling environment for integrating renewable electricity with farming.

**[Watch Video on YouTube: ▶ https://www.youtube.com/embed/Xx_75KImjkk ]**

## UPSC Civil Services Examination Previous Year Question (PYQ)

### *Prelims*

**Q. With reference to solar power production in India, consider the following statements: (2018)**

1. India is the third largest in the world in the manufacture of silicon wafers used in photovoltaic units.
2. The solar power tariffs are determined by the Solar Energy Corporation of India.

**Which of the statements given above is/are correct?**

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**Ans: (d)**

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### *Mains*

**Q.** India has immense potential of solar energy though there are regional variations in its developments. Elaborate **(2020)**

# Swachh Survekshan Awards 2025-26

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**Source: PIB**

## Why in News?

The President of India conferred the **Swachh Survekshan Awards 2025-26** at Vigyan Bhavan, New Delhi. In a significant shift in urban sanitation rankings, **Navi Mumbai (Maharashtra)** was recognised as **India's cleanest city,** securing the **top position in the Super Swachh League** .

## What are the Key Highlights of Swachh Survekshan 2025-26?

- **Theme:** The 10 th edition of the survey was conducted under the theme, '*Swachhata Ki Nayi Pehel-Badhayein Haath, Karein Safai Saath*' .
- **Top Cleanest Cities (Super Swachh League): Navi Mumbai** secured the 1 st position. **Surat** (Gujarat) and **Vijayawada** (Andhra Pradesh) finished 2 nd and 3 rd , respectively.
- **Population-wise Rankings:**
  - **3 to 10 Lakh Population:** Ujjain (Madhya Pradesh) ranked first, followed by Chandigarh and Guntur (Andhra Pradesh).
  - **Over 10 Lakh Population:** Greater Visakhapatnam (Andhra Pradesh) ranked first, followed by Jabalpur (Madhya Pradesh) and Lucknow.
- **Special Categories Introduced:**
  - **Clean Himalayan Cities:** Rudrapur (Uttarakhand) took the top spot, followed by Jammu and Shopian (J&K).
  - **Swachh River Front Towns:** Rewa (Madhya Pradesh) was ranked first, followed by Bilaspur (Chhattisgarh) and Varanasi (Uttar Pradesh).
  - **Swachh Coastal Towns:** Puri (Odisha) secured the first rank, with Alappuzha (Kerala) and Ratnagiri (Maharashtra) following closely.
  - **Best 'Swachh Sheher Jodi':** Rooted in India's timeless ***Guru-Shishya* tradition** , this initiative pairs a **high-performing "Mentor" city with a struggling "Mentee" city.** The goal is ' **Each One - Teach One'** , enabling cities to share innovations and best practices.
    - Under the initiative, 70 mentor cities were paired with 191 mentee cities to promote knowledge-sharing and improve urban sanitation outcomes.
    - Rankings are based on the combined score of the Mentor City and its paired Mentee City, rewarding joint performance and actual mentorship outcomes.
    - **Top Three Mentor-Mentee Pairs:** Ahmedabad (Mentor)–Sanand (Mentee), Indore (Mentor)–Depalpur (Mentee) and Tirupati (Mentor)–Kuppam (Mentee).
- **Improvement in Rankings:** Over 350 Urban Local Bodies (ULBs) improved by more than 500 positions,

while **1,700+ ULBs with populations below 50,000 improved their rankings compared with 2024–25.**




## Factors Driving Navi Mumbai's Success

Navi Mumbai, which was placed in the elite 'Super Swachh League' for consistently ranking in the top three over several years, finally secured the crown through:

- **Strict Enforcement:** Rigorous implementation of **solid waste management rules** by the Navi Mumbai Municipal Corporation (NMMC).
- **Continuous Maintenance:** 24/7 cleaning of roadsides and skywalks, alongside targeted sanitation drives in high-footfall public areas.
- **Citizen Participation:** Robust utilization of citizen feedback via the Swachhata app and the active involvement of civic employees and local representatives.

# Swachh Survekshan

- **Nodal Ministry:** Conducted annually by the **Ministry of Housing and Urban Affairs (MoHUA)** .
- **Scale and Scope:** Launched in 2016 with just 73 cities, it has grown into the world's largest urban cleanliness survey, covering **4,751 cities** in the 2025-26 edition and incorporating feedback from 2.7 crore citizens.
- **Assessment Framework** : The survey evaluated cities through 43 indicators across 10 assessment sections, with a maximum score of 12,500 marks, comprising 10,500 marks for on-ground assessment (including citizen feedback) and 2,000 marks for certification.
  - It assesses visible cleanliness, **waste segregation, solid waste processing, wastewater management,** mechanised desludging, sanitation workers' welfare and citizen grievance redressal.
- **Continuous Monitoring:** **Urban Local Bodies (ULBs)** must track sustained progress through monthly Management Information System (MIS) updates, with final rankings validated by ongoing citizen feedback.
- **Key Objectives of Swachh Survekshan:**
  - **Mission Acceleration:** Fast-track sanitation and cleanliness goals across cities.
  - **Healthy Competition:** Drive competitive improvements in urban waste management.
  - **Citizen Engagement:** Promote mass participation and awareness about *Swachhata* .
  - **Service Delivery:** Enhance the quality of sanitation services provided to residents.

# Frequently Asked Questions (FAQs)

**1. Which ministry conducts Swachh Survekshan, and when was it launched?**
It is conducted annually by the Ministry of Housing and Urban Affairs (MoHUA) and was launched in 2016 with 73 cities.

**2. Which city topped Swachh Survekshan 2025-26?**
Navi Mumbai (Maharashtra) secured the top position, followed by Surat (Gujarat) and Vijayawada (Andhra Pradesh).

**3. What is the Swachh Sheher Jodi initiative?**
It follows a Mentor-Mentee model, pairing high-performing cities with weaker-performing cities to facilitate knowledge-sharing and improve urban sanitation through peer learning.

**4. What are the major parameters assessed under Swachh Survekshan?**
Key parameters include visible cleanliness, waste segregation, wastewater treatment, mechanised desludging, citizen feedback and grievance redressal.

**5. What is the significance of Swachh Survekshan for urban governance?**
It promotes competitive federalism at the urban level, citizen participation, accountability and improved sanitation service delivery through performance-based rankings.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/VPf3dx3hB1I** ]

## UPSC Civil Services Examination, Previous Year Question (PYQ)

**Q. As per the Solid Waste Management Rules, 2016 in India, which one of the following statements is correct? (2019)**

**(a)** Waste generator has to segregate waste into five categories.
**(b)** The Rules are applicable to notified urban local bodies, notified towns and all industrial townships only
**(c)** The Rules provide for exact and elaborate criteria for the identification of sites for landfills and waste processing facilities.
**(d)** It is mandatory on the part of the waste generator that the waste generated in one district cannot be moved to another district.

**Ans: (c)**

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## *Mains:*

**Q.** What are the impediments in disposing of  huge quantities of discarded solid waste which are continuously being generated? How do we remove safely the toxic wastes that have been accumulating in our habitable environment? **(2018)**

# Maharashtra Beaches Receive 1st Blue Flag Certification

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**Source: TH**

**Three beaches in Maharashtra—Ladghar and Guhagar in Ratnagiri district and Shrivardhan in Raigad district** —have received international **Blue Flag certification** for 2026–27, marking the state's first entry into the globally recognised eco-label programme.

- **National Status:** India has **22 Blue Flag-certified beaches for 2026–27** , including **13 renewed certifications and nine newly certified beaches** , of which three are in Maharashtra.
- **Certification Standards:** The beaches met prescribed international standards for **water quality, environmental management, biodiversity conservation, cleanliness, tourist safety and sustainable tourism** .

## Blue Flag Certification

- **About:** An internationally recognised **voluntary eco-label awarded to beaches, marinas and sustainable tourism boats that meet prescribed environmental, educational, safety and management standards** .
- **Awarding Organisation:** Administered by the **Foundation for Environmental Education (FEE),** an international non-profit organisation headquartered in Copenhagen, Denmark.
- **Assessment:** Certification is awarded following evaluation through the Blue Flag programme's assessment process, including review by the **International Blue Flag Jury** .
- **Major Certification Criteria:** Blue Flag beach certification traditionally covers **four major categories:**
  - **Environmental Education and Information:** Public awareness, environmental information and educational activities.
  - **Water Quality:** Compliance with prescribed bathing-water quality standards.
  - **Environmental Management:** Waste management, sanitation and protection of coastal ecosystems.
  - **Safety and Services:** Lifeguards, emergency preparedness, accessibility and visitor facilities.
- **Validity and Monitoring:** Certification applies to a **specified season, and participating sites must maintain compliance** with applicable standards throughout the award period.
- **Indian Initiative:** The **Beach Environment & Aesthetics Management Services (BEAMS)** programme, under the **Integrated Coastal Zone Management (ICZM) framework** of the **Ministry of Environment, Forest and Climate Change (MoEFCC)** , promotes beach cleanliness, coastal conservation, visitor safety and sustainable beach management, supporting Blue Flag certification.

**Read More:** **Blue Flag Certification**

# Discovery of 5 New Marine Nematode Species in Kerala

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**Source:TH**

Researchers from the University of Kerala have discovered **five new species of free-living marine nematodes** from the mangrove sediments along the Kerala coast.

- **First Record in the Region:** This discovery represents the **first confirmed record of the nematode genus *Pseudochromadora* from the Arabian Sea** .
- **Newly Identified Species:** The five species new to science ( ***Pseudochromadora coronate* , *P. cryptica* , *P. kairali* , *P. kandeli* , and *P. longicaudata*** ) were collected from the **mangrove forests of Kollam and Kannur** .
- **Ecological Role:** These marine nematodes are **free-living (non-parasitic)** microscopic roundworms.
  - Despite their minuscule size, they are vital for **organic matter decomposition and nutrient cycling** in marine and estuarine habitats.
- **Meiofaunal Biodiversity:** The study significantly reduces the knowledge gap regarding the highly diverse but poorly documented **meiofaunal communities (minute animals living in soil and aquatic sediments) of the Indian Ocean** .
  - The findings emphasize that **mangrove conservation must safeguard the entire biological community** , including microscopic species that are foundational to sustaining essential ecosystem processes.

## Nematodes

- **About:** Nematodes, commonly known as roundworms, belong to the phylum Nematoda.
  - They are unsegmented, cylindrical worms with a **fluid-filled body cavity (pseudocoelom)** and are among the most abundant multicellular animals on Earth.
- **Habitat:** They are incredibly adaptable. They exist in almost every environment from marine and freshwater ecosystems to soils, deserts, and the ocean floor.
- **Classification:**
  - **Free-living Nematodes:** Feed on bacteria, fungi, algae, or other microscopic organisms (e.g., *Corononema dhriti* ).
  - **Parasitic Nematodes:** Infect plants, animals, and humans, often causing diseases (e.g., Root-knot nematodes in agriculture, Ascaris in humans).
  - **Entomopathogenic Nematodes (EPNs):** A specialised group of **insect-parasitic nematodes** that typically use **symbiotic bacteria** to kill insect hosts, making them useful in biological pest control.

**Read more:** **2025 Animal and Plant Discoveries and 111th Foundation Day of ZSI**

# SS United States Sunk to Create World's Largest Artificial Reef

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**Source: TOI**

The **historic SS United States, once the world's fastest transatlantic passenger liner** , was deliberately sunk in the **Gulf of Mexico** off Florida, USA, to create the **world's largest ship-based artificial reef ,** aimed at supporting marine biodiversity and diving tourism.

- **About SS United States:** Launched in **1951** , the **990-foot-long ocean liner** , designed by American naval architect **William Francis Gibbs** , was once the world's fastest transatlantic passenger liner.
- **Artificial Reef Creation:** The vessel was deliberately sunk off **Destin, Florida, in the Gulf of Mexico** , to create the **world's largest ship-based artificial reef** , following the removal of hazardous materials and pollutants to minimise environmental risks.
- **Previous Record Holder:** The **USS Oriskany** , an **888-foot-long former US aircraft carrier** sunk off Pensacola, Florida, in **2006** , previously held the record for the world's largest ship-based artificial reef.

## Artificial Reefs

- **About:** Artificial reefs are **human-made underwater structures** deliberately placed on the seabed to provide habitats for marine organisms and support ecological or recreational activities.
- **Materials:** They may be constructed using **concrete modules, specially designed reef structures or decommissioned ships** , following appropriate environmental preparation.
- **Ecological Benefits:** Artificial reefs provide surfaces for the colonisation of **corals, algae and other marine organisms** , increasing habitat complexity and potentially supporting fish populations.
  - However, their long-term ecological benefits depend on suitable materials, environmental conditions and effective monitoring.
- **Economic Benefits:** Artificial reefs can promote **diving tourism, recreational fisheries and coastal livelihoods** , contributing to local economic development.

**Read More:** World's Largest Deep Sea Coral Reef